

Customer Shopping Behavior Analysis

Turning 3,900 transaction records into a revenue and retention strategy

Python (Pandas)

Data cleaning & feature engineering

SQL (Oracle)

10 core business questions answered

Power BI

Interactive executive dashboard

Kaggle Dataset

Customer Shopping Trends — 3,900 records

Business Context

The Retailer

A mid-size fashion retailer selling Clothing, Footwear, Accessories, and Outerwear across the US — through both online channels and a mix of subscription and one-off purchase models.

The Gap

Leadership had transaction-level data but no consolidated view of **who spends the most, which products convert best, and whether discounts and subscriptions are actually driving revenue.**

The Brief

This was a self-directed project framed as a real retail analytics engagement. The mandate: turn raw transactional data into decisions the merchandising and marketing teams could act on — without requiring them to write a single query.

Deliverables

- Cleaned, structured dataset loaded into a live Oracle database
- 10 SQL queries answering core business questions
- Interactive Power BI dashboard for stakeholder self-service
- 3–4 concrete, data-backed business recommendations

Objective & Scope

Project Objectives

01

Clean & Structure

Prepare raw transactional data for reliable analysis — impute, standardize, and enrich before any querying begins.

02

Answer 10 Business Questions

Write SQL queries that address real merchandising and marketing decisions — not just descriptive stats.

03

Build a Self-Serve Dashboard

Deliver an interactive Power BI dashboard so stakeholders can explore cuts without requesting new queries.

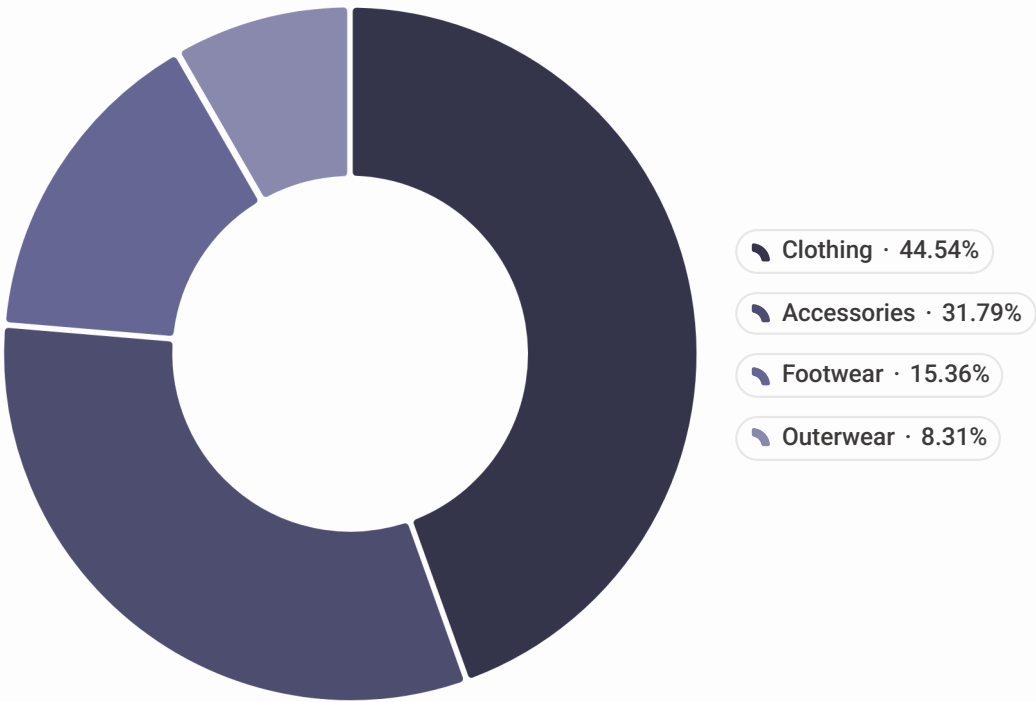
04

Translate to Recommendations

Distill findings into 3–4 concrete actions the business can take in the next planning cycle.

Dataset at a Glance

Attribute	Detail
Source	Kaggle — Customer Shopping Trends
Records	3,900 customer transactions
Attributes	18 (demographics, product, price, ratings, shipping, discounts, subscription, frequency)
Clothing	1,737 orders — 44.5%
Accessories	1,240 orders — 31.8%
Footwear	599 orders — 15.4%
Outerwear	324 orders — 8.3%



Step 1 — Data Cleaning & Feature Engineering (Python)

Raw data isn't analysis-ready. Before writing a single SQL query, the dataset was cleaned and enriched in Pandas.

	CUSTOMER_ID	AGE	GENDER	ITEM_PURCHASED	CATEGORY	PURCHASE_AMOUNT	LOCATION	size	COLOR	SEASON	REVIEW_RATING	SUBSCRIPTION_STATUS	SHIP
1	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express
2	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express
3	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free
4	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next
5	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free
6	6	46	Male	Sneakers	Footwear	20	Wyoming	M	White	Summer	2.9	Yes	Stand
7	7	63	Male	Shirt	Clothing	85	Montana	M	Gray	Fall	3.2	Yes	Free
8	8	27	Male	Shorts	Clothing	34	Louisiana	L	Charcoal	Winter	3.2	Yes	Free
9	9	26	Male	Coat	Outerwear	97	West Virginia	L	Silver	Summer	2.6	Yes	Express
10	10	57	Male	Handbag	Accessories	31	Missouri	M	Pink	Spring	4.8	Yes	2-Day
11	11	53	Male	Shoes	Footwear	34	Arkansas	L	Purple	Fall	4.1	Yes	Store
12	12	30	Male	Shorts	Clothing	68	Hawaii	S	Olive	Winter	4.9	Yes	Store
13	13	61	Male	Coat	Outerwear	72	Delaware	M	Gold	Winter	4.5	Yes	Express
14	14	65	Male	Dress	Clothing	51	New Hampshire	M	Violet	Spring	4.7	Yes	Express
15	15	64	Male	Coat	Outerwear	53	New York	L	Teal	Winter	4.7	Yes	Free
16	16	64	Male	Skirt	Clothing	81	Rhode Island	M	Teal	Winter	2.8	Yes	Store
17	17	25	Male	Sunglasses	Accessories	36	Alabama	S	Gray	Spring	4.1	Yes	Next
18	18	53	Male	Dress	Clothing	38	Mississippi	XL	Lavender	Winter	4.7	Yes	2-Day
19	19	52	Male	Sweater	Clothing	48	Montana	S	Black	Summer	4.6	Yes	Free
20	20	66	Male	Pants	Clothing	90	Rhode Island	M	Green	Summer	3.3	Yes	Stand
21	21	21	Male	Pants	Clothing	51	Louisiana	M	Black	Winter	2.8	Yes	Express
22	22	31	Male	Pants	Clothing	62	North Carolina	M	Charcoal	Winter	4.1	Yes	Store
23	23	56	Male	Pants	Clothing	37	California	M	Peach	Summer	3.2	Yes	Store

Missing Ratings

Imputed missing *Review Rating* values using the **median rating per category** — not a global median — to prevent footwear ratings from skewing clothing ratings and vice versa.

Standardized Schema

Lower-cased and snake-cased all column names for consistent SQL querying. Example: *Purchase Amount (USD)* → *purchase_amount*.

Age Segmentation

Created an *age_group* feature using quartile binning (*pd.qcut*): Young Adult (≤31), Adult (32–44), Middle-aged (45–57), Senior (58+) — each group represents a real quarter of the customer base.

Frequency → Numeric

Mapped categorical frequency labels (Weekly, Fortnightly, Monthly, Quarterly, Annually, etc.) to actual day intervals — making frequency usable in calculations, not just a label.

Redundancy Check

Verified *discount_applied* and *promo_code_used* were 100% identical columns and dropped the duplicate — preventing any dashboard from double-counting a discount signal.

Loaded to Oracle DB

Pushed the cleaned dataset into an Oracle database via SQLAlchemy, so the SQL layer worked against a real relational source rather than a flat file.

Step 2 — SQL Analysis: Questions 1–5

Each query was written to answer a question a merchandising or marketing manager would actually ask.

Q1 — Revenue Split by Gender

	GENDER	REVENUE
1	Male	157890
2	Female	75191

Male: **\$157,890 (67.7%)** · Female: **\$75,191 (32.3%)**. Despite female customers being close to a third of order volume, average order values are comparable across gender — the revenue gap is volume-driven, not spend-driven.

Q2 — High-Value Discount Users

CUSTOMER_ID	PURCHASE_AMOUNT
1	2
2	3
3	4
4	7
5	9
6	12
7	13
8	16
9	20
10	22
11	24
12	29
13	32
14	33
15	35
16	37
17	40
18	41
19	43
20	44
21	55
22	57
23	58
24	60

Isolated customers who redeemed a discount *and* still spent above the \$59.75 average order value — price-sensitive but high-intent shoppers, a distinct segment from bargain-only buyers.

Q3 — Top 5 Highest-Rated Products

	ITEM_PURCHASED	AVG_RATING
1	Gloves	3.9
2	Sweater	3.8
3	Handbag	3.8
4	Sneakers	3.8
5	Sandals	3.8

Gloves (3.86), Sandals (3.84), Boots (3.82), Hats (3.80), Skirts (3.79) — accessories and footwear outperform core clothing items on customer satisfaction.

Q4 — Standard vs Express Shipping AOV

	SHIPPING_TYPE	AVG_AMOUNT
1	Express	60.48
2	Standard	58.46

Express: **\$60.48** vs Standard: **\$58.46**. Customers who pay for faster shipping also spend ~3.5% more per order — shipping speed is a viable upsell lever, not just a cost center.

Q5 — Do Subscribers Spend More?

SUBSCRIPTION_STATUS	TOTAL_CUSTOMERS	AVG_SPEND	TOTAL_REVENUE
Pin (Ctrl+I)	2847	59.87	170436
2 Yes	1053	59.49	62645

Counter-intuitively, **non-subscribers spend slightly more per order** (\$59.87 vs \$59.49) and generate far more total revenue (\$170,436 vs \$62,645) — simply because they are 2.7× more numerous. Subscription status is not currently a strong spend predictor; it is a volume/loyalty lever, not a spend lever.

Step 2 — SQL Analysis: Questions 6–10

Q6 — Highest Discount Penetration

	ITEM_PURCHASED	DISCOUNT_RATE
1	Hat	50
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

Hats (50.0%), Sneakers (49.7%), Coats (49.1%), Sweaters (48.2%), Pants (47.4%) — nearly half of all orders in these categories involve a discount. A margin-risk signal worth a pricing review before the next promo cycle.

Q7 — Customer Segmentation by Purchase History

	CUSTOMER_SEGMENT	Number of Customers
1	Loyal	3116
2	Returning	701
3	New	83

Loyal (11+ purchases): **3,116 customers (79.9%)** · Returning (2–10): **701 (18.0%)** · New (1 purchase): **83 (2.1%)**. The base is overwhelmingly loyalty-driven — reframing the real priority as retention economics, not acquisition.

Q8 — Top 3 Products per Category

	ITEM_RANK	CATEGORY	ITEM_PURCHASED	TOTAL_ORDERS
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

Used a window function (*ROW_NUMBER() OVER PARTITION BY*) to rank best-sellers within each category individually — avoiding a global top-seller list dominated by the largest category (Clothing).

Q9 — Do Repeat Buyers Subscribe?

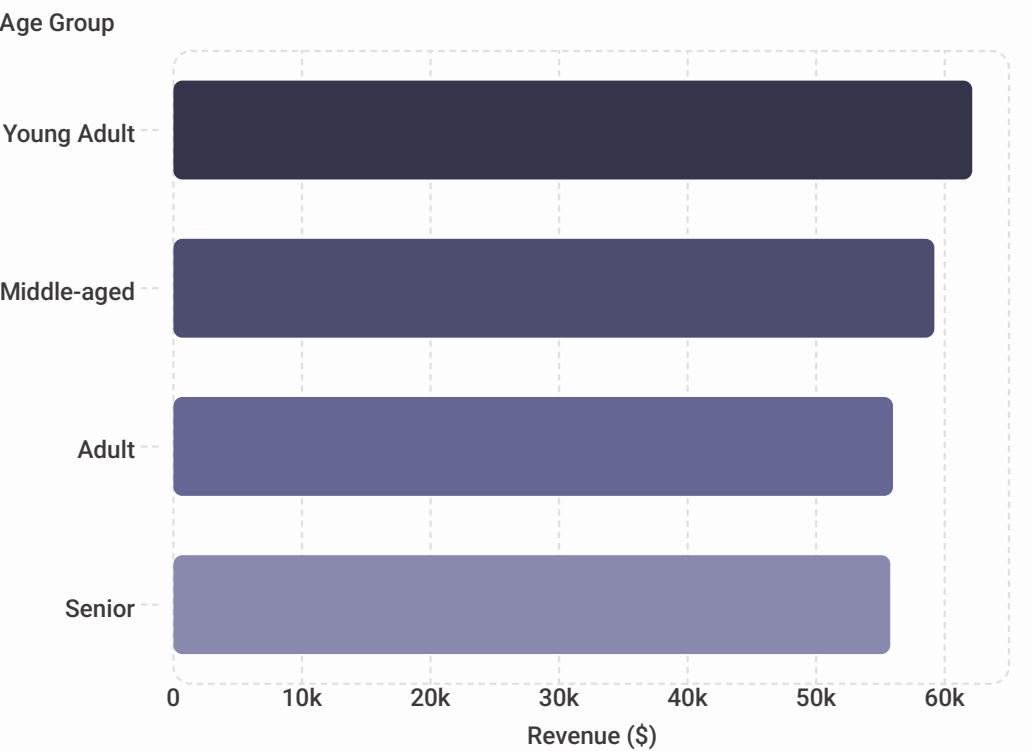
	SUBSCRIPTION_STATUS	REPEAT_BUYERS
1	No	2518
2	Yes	958

Of repeat buyers (5+ purchases), only **958 subscribe vs 2,518 who don't**. Loyalty is being earned through product and experience — not the subscription program. The subscription offer isn't currently converting the retailer's best customers.

Q10 — Revenue by Age Group

	AGE_GROUP	TOTAL_REVENUE
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

Young Adults lead at **\$62,143**, followed by Middle-aged (\$59,197), Adult (\$55,978), and Senior (\$55,763). Revenue is fairly evenly spread across age bands — marketing should not over-index on any single generation.



Step 3 — Power BI Dashboard

Built an interactive executive dashboard so stakeholders can filter and explore without touching SQL.



3.9K

Customers

\$233K

Total Revenue

\$59.76

Avg Order Value

3.75

Avg Rating

Dashboard Visuals

- **Sales by Category** (donut) — Clothing 44.7%, Accessories 31.8%, Footwear 15.5%, Outerwear 8.0%
- **Sales by Gender** (pie) — visual confirmation of the 67.7% / 32.3% split from Q1
- **Sales by State** (map) — geographic concentration for regional campaign targeting
- **Sales by Size & Category** (streamgraph) — Medium and Large dominate across categories
- **Sales by Item Name** (treemap) — Blouse, Shirt, Dress, Pants lead individual SKU revenue
- **Top 5 Color by Sales** — combo chart pairing sales volume with average rating per color, to catch colors that sell well but rate poorly (a restock/quality flag)

Slicer Coverage

Stakeholders can self-serve any cut of the data without requesting a new query. Slicers include:

- Discount · Frequency · Product · Payment
- Size · Location · Category · Color
- Gender · Season · Subscription · Shipping

☐ Every slicer combination updates all visuals simultaneously — enabling merchandising and marketing teams to explore their own hypotheses without analyst support.

Key Recommendations

1. Redesign the Subscription Program

The subscription program does not correlate with higher order values and fails to capture the most loyal (5+ purchase) customers — only 958 of 3,476 repeat buyers subscribe. Redesign the offer around perks loyal customers actually value, or reposition it purely as a retention and communication channel rather than a spend driver.

2. Review Discount Depth on High-Penetration Lines

Hats, Sneakers, Coats, Sweaters, and Pants all carry near-50% discount penetration rates. This is a margin-risk signal worth investigating before the next promotional cycle. Consider testing reduced discount depth or time-limited offers to protect margin without sacrificing conversion.

3. Use Express Shipping as an Upsell Lever

Express shipping customers spend ~3.5% more per order (\$60.48 vs \$58.46 standard). Bundling or promoting faster shipping at checkout — through a threshold offer or loyalty perk — has measurable upside and positions shipping as a revenue driver rather than a pure cost center.

4. Prioritize Retention Over Acquisition

With ~80% of customers already in the "Loyal" segment (11+ purchases), broad top-of-funnel acquisition spend has limited incremental impact. Marketing budget is better directed at deepening loyal-customer value through cross-sell, bundles, and category expansion — particularly into high-satisfaction lines like Accessories and Footwear.

Project Links

Kaggle Dataset

Customer Shopping Behavior – Cleaned Dataset

The source dataset used for this analysis – 3,900 transactions, 18 attributes, published on Kaggle.

Kaggle Notebook

Customer Shopping Behavior Analysis

Full Python notebook covering data cleaning, feature engineering, and exploratory analysis in Pandas.

GitHub Repository

SQL + README

All 10 SQL queries with documentation, the cleaned dataset schema, and project README for reproducibility.